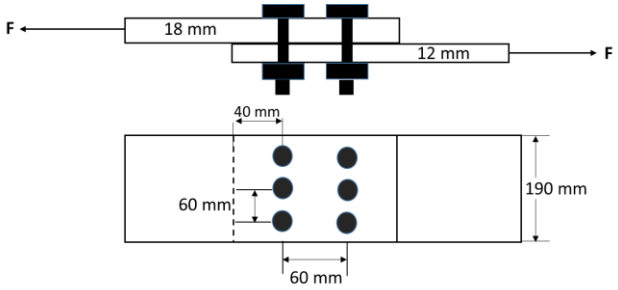
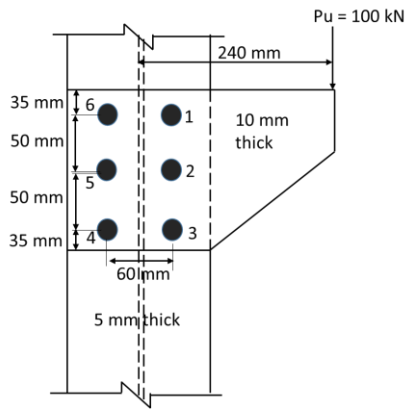
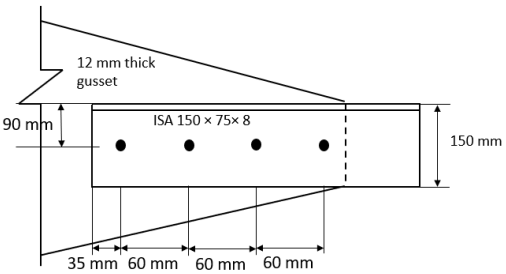
MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OF CIVIL ENGINEERING			
END SEMESTER EXAMINATION		**NOVEMBER, 2024**	
Level	UG	**Curriculum**	2022
Program Name	Civil engineering	**Course Type**	C
Course Name	Basic Structural Steel Design	**Course Code**	CIE 3121
Semester	V	**Max Marks**	50
Date	19-11-2024	**Duration**	3 Hrs.

Instructions:

- **Answer ALL the questions.**
- **Missing data if any may be suitably assumed.**
- **Assume Fe410 steel with f_y = 250 N/mm^2 unless specified otherwise.**
- **Use of IS800: 2007 and SP6 is permitted.**

Q. No.	Question	Marks	CO	PO	BL
1A	Determine the maximum load carrying capacity of a lap connection shown in the figure. The bolts are M22 of grade 4.6 	5	1	1,2, 6,8, 10, 12	3
1B	Determine the maximum force experienced by bolts 1 and 2 in the connection shown in figure. The bolts are M20 of grade 4.6. 	5	1	1,2, 6,8, 10, 12	3

2A	Determine the block shear capacity of the connection shown in figure. The bolts used are M16 of 4.6 grade 	5	2	1,2, 6,8, 10, 12	3
2B	Determine the design compressive strength of a column of 3.25 m length made up of ISMB 175 @ 19.3 kg/m. The one end of the column is fixed, and the other end is restrained against rotation but free in translation.	5	3	1,2, 6,8, 10, 12	3
3A	Design a single angle strut connected to a gusset plate using i) weld (k = 0.7) ii) 2 bolts (k = 0.85). The length of the strut is 2.5 m and carries a factored axial force of 300 kN. Assume f_{cd} = 100 N/mm^2 for initial area calculation.	5	3	1,2, 6,8, 10, 12	3
3B	It is required to splice two ISMB 550 @ 103.7 kg/m to transmit a factored axial load of 2000 kN and a factored moment of 100 kN-m. Determine the size of column splice only. The ends of the columns are milled/machine finished.	5	4	1,2, 6,8, 10, 12	3
4A	Classify the following sections based on the limiting width to thickness ratio, i) ISLB 150@ 14.2 kg/m ii) ISA 40×40×6	5	5	1,2, 6,8, 10, 12	3
4B	Design a laterally supported beam of span 4.20 m to carry a factored load of 20 kN/m. The beam is simply supported. Check for deflection, web buckling and crippling need not be carried out.	5	5	1,2, 6,8, 10, 12	3
5A	Determine the safe working load that ISMB 225 @ 31.2 kg/m can carry. The beam is simply supported over a span 4 m. The reinforced concrete floor is capable of providing lateral restrain to the compression flange of the beam. Assume $V_u = 0.5\ V_d$.	5	5	1,2, 6,8, 10, 12	3
5B	Check ISMB 500 @ 86.9 kg/m for web buckling and crippling strength near supports. The stiff bearing length is 100 mm. The factored shear force on the beam is 115 kN.	5	5	1,2, 6,8, 10, 12	3

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO	Total Marks
1	10
2	5
3	10
4	5
5	20

BL	Total Marks
2	**-**
3	**50**
4	**-**
5	**-**